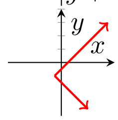
MTH 137(B) Dr. James Ashe
Exam 2 Wednesday 13^{th} March, 2013 **Name:**

You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

Determine whether the following relation defines y as a function of x.

1. $x = |y+3| - 1$



Find the domain and range of the relation. Determine if the relation is a function.

2. $g = \{(2,7),(30,-4),(45,7),(2,5)\}$

a. Domain:

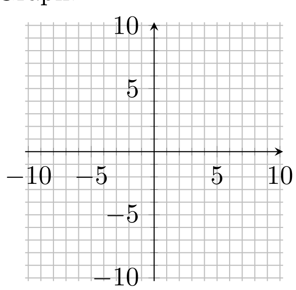
b. Range:

c. Is g a function?

Sketch the graph of the function, then give its domain and range.

3. $y = -2x + 4$

a. Graph:

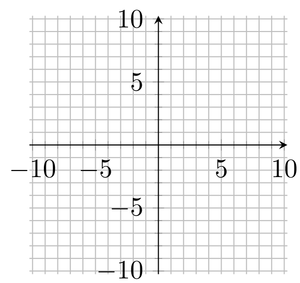


b. Domain:

c. Range:

4. $f(x) = \begin{cases} x+2 & \text{for } x > 1 \\ -x-4 & \text{for } x \leq 1 \end{cases}$

a. Graph:



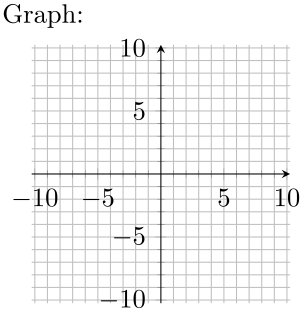
b. Domain:

c. Range:

Use transformation to graph the function. State the domain and range using interval notation.

5. $f(x) = -(x+3)^2 + 5$

a. Graph:

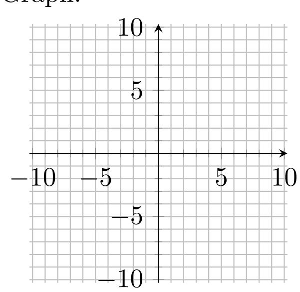


b. Domain:

c. Range:

6. $g(x) = -\frac{1}{2}|x-1|$

a. Graph:

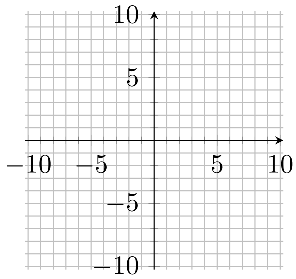


b. Is $g(x)$ one-to-one? i.e., does it have an inverse?

Sketch the graph and state the domain and range. Identify any intervals on which $f(x)$ is increasing or decreasing.

7. $f(x) = -\sqrt{x-3}$

a. Graph:



b. Increasing:

c. Decreasing:

Use the function $h(x) = 4x + 5$ to complete the following problems.

8. Find $h(4)$